

3 Introduction to Infinite Series

In the previous chapter, we studied *sequences*, that is, ordered infinite collections of real numbers. A natural question is:

Can we meaningfully add infinitely many real numbers?

For a finite collection of real numbers $a_1, a_2, \dots, a_n$, the sum

$$a_1 + a_2 + \dots + a_n$$

is well defined. It is therefore natural to consider the infinite expression

$$\sum_{n=1}^{\infty} a_n.$$

Unlike finite sums, however, infinite sums cannot be manipulated arbitrarily. Without a precise definition, they may lead to contradictions. For example, consider the series

$$1 - 1 + 1 - 1 + \dots.$$

If we group the terms as

$$(1 - 1) + (1 - 1) + (1 - 1) + \dots,$$

we obtain

$$0 + 0 + 0 + \dots = 0.$$

But if we group them as

$$1 + (-1 + 1) + (-1 + 1) + \dots,$$

we obtain

$$1 + 0 + 0 + \dots = 1.$$

To avoid such contradictions, we need a precise definition of the sum of an infinite series. Developing this definition and studying its properties is the main objective of this chapter.

3.1 Convergence and sum of an infinite series

Definition 3.1. Let (a_n) be a sequence of real numbers. A formal expression of the form

$$\sum_{n=1}^{\infty} a_n,$$

or simply $\sum a_n$ or $\sum_n a_n$ is called an ***infinite series***.

Definition 3.2 (Partial Sum). Let $\sum_{n=1}^{\infty} a_n$ be an infinite series. For each $n \in \mathbb{N}$, the finite sum

$$s_n := a_1 + a_2 + \dots + a_n$$

is called the n th ***partial sum*** of the series.

Remark 3.1. Associated with the series $\sum_{n=1}^{\infty} a_n$, we have the sequence of partial sums (s_n) , where

$$s_n = a_1 + a_2 + \cdots + a_n, \quad n \in \mathbb{N}.$$

Thus, every infinite series naturally determines a sequence of partial sums.

Definition 3.3 (Convergent Series). An infinite series $\sum_{n=1}^{\infty} a_n$ is said to be **convergent** if the sequence of partial sums (s_n) is convergent.

Remark 3.2. If $s_n \rightarrow s$, then we call s the **sum** of the series and write $\sum_{n=1}^{\infty} a_n = s$.

Example 3.3. 1. Consider the series $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \cdots$. Let $\sum_{n=1}^{\infty} a_n$, where $a_n = \frac{1}{n(n+1)}$. If $s_n = u_1 + u_2 + \cdots + a_n$, then

$$\begin{aligned} s_n &= \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} \\ &= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \\ &= 1 - \frac{1}{n+1}. \end{aligned}$$

Hence, $\lim_{n \rightarrow \infty} s_n = 1$. Therefore, the series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ is convergent and its sum is 1.

2. (**Geometric Series**) Let $a \in \mathbb{R}$ satisfy $|a| < 1$. Consider the infinite series $\sum_{n=0}^{\infty} a^n$. Since the n th partial sum is

$$s_n = \sum_{k=0}^n a^k = \frac{1 - a^{n+1}}{1 - a},$$

and $a^{n+1} \rightarrow 0$, it follows that $s_n \rightarrow \frac{1}{1-a}$. Thus, $\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}$, $|a| < 1$.

Definition 3.4 (Divergent Series). An infinite series $\sum_{n=1}^{\infty} a_n$ is said to be **divergent** if the sequence of partial sums (s_n) is divergent.

Example 3.4. 1. Consider the series $1 + 2 + 3 + \cdots$. Its sequence of partial sums is

$$s_n = 1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Since $\lim_{n \rightarrow \infty} s_n = \infty$, the sequence of partial sums does not converge. Hence, the series

$\sum_{n=1}^{\infty} n$ is divergent.

2. (**Harmonic Series**) Consider the series

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots.$$

Let $\sum_{n=1}^{\infty} a_n$ be the series, where $a_n = \frac{1}{n}$. Let $s_n = a_1 + a_2 + \cdots + a_n$ be the sequence of partial sums. Then

$$s_2 = 1 + \frac{1}{2},$$

$$s_4 = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} > 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) = 1 + 2 \cdot \frac{1}{2},$$

$$s_8 = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) > 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) = 1 + 3 \cdot \frac{1}{2},$$

$$s_{16} > 1 + 4 \cdot \frac{1}{2},$$

$\vdots$

and, in general,

$$s_{2^n} > 1 + n \cdot \frac{1}{2}.$$

Hence, $\lim_{n \rightarrow \infty} s_{2^n} = \infty$. Since $s_{n+1} - s_n = a_{n+1} > 0$ for all $n \in \mathbb{N}$, the sequence $\{s_n\}$ is monotone increasing. As the subsequence $\{s_{2^n}\}$ is unbounded, the sequence $\{s_n\}$ is also unbounded. Therefore, $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent.

3. (**p-Series**) For $p > 0$, consider the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$. Then

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

is convergent if and only if $p > 1$. Equivalently,

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \begin{cases} \infty, & 0 < p \leq 1, \\ \text{converges}, & p > 1. \end{cases}$$

In particular, the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent, whereas the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent.

Theorem 3.5. Let m be a natural number. Then the two series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=m+1}^{\infty} a_n$ converge or diverge together.

Proof. Let $s_n = a_1 + a_2 + \cdots + a_n$, and $t_n = a_{m+1} + a_{m+2} + \cdots + a_{m+n}$. Then $t_n = s_{m+n} - s_m$, where s_m is a fixed number. If $\{s_n\}$ converges, then $\{t_n\}$ converges, and conversely. If $\{s_n\}$ diverges, then $\{t_n\}$ diverges, and conversely. Hence, both the sequences $\{s_n\}$ and $\{t_n\}$, and consequently the series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=m+1}^{\infty} a_n$, converge or diverge together. $\square$

Remark 3.6. The theorem states that removing a finite number of terms from the beginning of a series, or adding a finite number of terms to the beginning of a series, does not affect its convergence or divergence.

Theorem 3.7 (Algebra of Convergent Series). Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be two convergent series with sums A and B , respectively. Then:

1. The series $\sum_{n=1}^{\infty} (a_n + b_n)$ is convergent and

$$\sum_{n=1}^{\infty} (a_n + b_n) = A + B.$$

2. For every $\lambda \in \mathbb{R}$, the series $\sum_{n=1}^{\infty} \lambda a_n$ is convergent and

$$\sum_{n=1}^{\infty} \lambda a_n = \lambda A.$$

Theorem 3.8 (Cauchy Criterion). The series $\sum_{n=1}^{\infty} a_n$ converges if and only if for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$n, m \geq N \Rightarrow |s_n - s_m| < \epsilon,$$

where (s_n) denotes the sequence of partial sums. Thus, the series $\sum_{n=1}^{\infty} a_n$ converges if and only if for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$n > m \geq N \Rightarrow |a_{m+1} + a_{m+2} + \cdots + a_n| < \epsilon.$$

Proof. The series $\sum_{n=1}^{\infty} a_n$ converges if and only if the sequence (s_n) of partial sums converges. By the Cauchy Criterion for sequences, this is equivalent to saying that (s_n) is a Cauchy sequence. Since

$$s_n - s_m = a_{m+1} + a_{m+2} + \cdots + a_n,$$

the second statement follows immediately. $\square$

Theorem 3.9. Suppose that the series $\sum_{n=1}^{\infty} a_n$ converges to s . Then, for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$\left| \sum_{n=N+1}^{\infty} a_n \right| < \epsilon.$$

Proof. Let $\epsilon > 0$ be given. Since $s_n \rightarrow s$, there exists $N \in \mathbb{N}$ such that $|s - s_N| < \epsilon$. Since $\sum_{n=N+1}^{\infty} a_n = s - s_N$, it follows that

$$\left| \sum_{n=N+1}^{\infty} a_n \right| = |s - s_N| < \epsilon.$$

This completes the proof. $\square$

Theorem 3.10. If the series $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n \rightarrow \infty} a_n = 0$.

Proof. Let $s_n = \sum_{k=1}^n a_k$ be the sequence of partial sums. Since the series converges, (s_n) converges. Hence, by the Cauchy Criterion for sequences, for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$|s_n - s_m| < \epsilon, \quad \forall n, m \geq N.$$

Now let $n \geq N + 1$. Then $n - 1 \geq N$, and therefore

$$|a_n| = |s_n - s_{n-1}| < \epsilon.$$

Hence $a_n \rightarrow 0$. □

Example 3.11. Show that the series $\sum_{n=1}^{\infty} a_n$ is divergent, where $a_n = \frac{n}{n+1}$.

Proof. Since $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{n}{n+1} = 1 \neq 0$, the necessary condition for the convergence of a series is violated. Hence $\sum_{n=1}^{\infty} a_n$ is divergent. □

Theorem 3.12. If the series $\sum_{n=1}^{\infty} |a_n|$ converges, then the series $\sum_{n=1}^{\infty} a_n$ also converges.

Proof. Let $s_n = \sum_{k=1}^n a_k$ and $\sigma_n = \sum_{k=1}^n |a_k|$ denote the sequences of partial sums of $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} |a_n|$, respectively.

Since (σ_n) converges, it is a Cauchy sequence. Thus, for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|\sigma_n - \sigma_m| < \epsilon, \quad \forall n > m \geq N.$$

Then

$$|s_n - s_m| = |a_{m+1} + \cdots + a_n| \leq |a_{m+1}| + \cdots + |a_n| = \sigma_n - \sigma_m < \epsilon.$$

Hence, (s_n) is a Cauchy sequence. By the Cauchy Criterion for sequences, (s_n) converges.

Therefore, $\sum_{n=1}^{\infty} a_n$ converges. □

Example 3.13. Show that the series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \sum_{n=1}^{\infty} a_n$, where $a_n = \frac{(-1)^{n+1}}{n}$, is convergent.

Proof. Let $s_n = a_1 + a_2 + \cdots + a_n$ be the sequence of partial sums. Then, for any $p \in \mathbb{N}$,

$$|s_{n+p} - s_n| = \left| \frac{1}{n+1} - \frac{1}{n+2} + \frac{1}{n+3} - \cdots + (-1)^{p-1} \frac{1}{n+p} \right| < \frac{1}{n+1}.$$

Let $\varepsilon > 0$. Then $|s_{n+p} - s_n| < \varepsilon$ whenever $n > \frac{1}{\varepsilon} - 1$. Choose $m = \left\lfloor \frac{1}{\varepsilon} - 1 \right\rfloor + 2$. Then m is a natural number and

$$|s_{n+p} - s_n| < \varepsilon$$

for all $n \geq m$ and $p = 1, 2, 3, \dots$. Hence $\{s_n\}$ is a Cauchy sequence. Since $\mathbb{R}$ is complete, $\{s_n\}$ converges. Therefore, $\sum_{n=1}^{\infty} a_n$ is convergent. □

Exercise 3.1. 1. Consider the constant sequence $a_n = c$, $n \in \mathbb{N}$, where $c \in \mathbb{R}$ is fixed. Determine whether the series $\sum_{n=1}^{\infty} a_n$ is convergent.

2. Find the sum of each of the following infinite series, if it exists.

$$(a) \sum_{n=1}^{\infty} \frac{1}{2^n + 2^{n+1}}.$$

$$(b) \sum_{n=1}^{\infty} \frac{3}{(3n-2)(3n+1)}.$$

3. Using the Cauchy criterion, determine whether each of the following series converges.

$$(a) \sum_{n=1}^{\infty} \frac{n}{2^n}.$$

$$(b) \sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)}.$$

3.2 Tests of convergence of a series of positive terms

The convergence or divergence of a series is determined by its sequence of partial sums. Since partial sums are often difficult to compute explicitly, we use *convergence tests*, which determine the nature of a series without evaluating its partial sums directly. We now study some of these tests.

Definition 3.5. A series $\sum_{n=1}^{\infty} a_n$ is called a series of positive terms if $a_n > 0$, for all $n \in \mathbb{N}$.

Theorem 3.14. A series of positive real numbers $\sum_{n=1}^{\infty} a_n$ is convergent if and only if the sequence of partial sums $\{s_n\}$, $s_n = \sum_{k=1}^n a_k$, is bounded above.

Remark 3.15. Since $s_n = a_1 + a_2 + \cdots + a_n$ is a monotone increasing sequence, if it is not bounded above, then $\lim_{n \rightarrow \infty} s_n = \infty$. Hence, the series $\sum_{n=1}^{\infty} a_n$ diverges to ∞ . Therefore, a series of positive real numbers either converges to a finite real number or diverges to ∞ .

Theorem 3.16 (Comparison Test). Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be two series of nonnegative real numbers. Suppose that $a_n \leq b_n$ for all $n \in \mathbb{N}$. Then:

1. If $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ also converges.
2. If $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} b_n$ also diverges.

Example 3.17. The series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent. Since

$$0 \leq \frac{1}{n^3} \leq \frac{1}{n^2}, \quad \forall n \in \mathbb{N},$$

the Comparison Test implies that $\sum_{n=1}^{\infty} \frac{1}{n^3}$ is also convergent.

Example 3.18. The harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent. Since

$$\frac{1}{n} \leq \frac{1}{\sqrt{n}}, \quad \forall n \in \mathbb{N},$$

the Comparison Test implies that $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ is also divergent.

Theorem 3.19 (Limit Form of the Comparison Test). Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be two series of positive real numbers. If

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = l,$$

where $0 < l < \infty$, then the two series either both converge or both diverge.

Example 3.20. Test the convergence of the series

$$\frac{1+2}{2^3} + \frac{1+2+3}{3^3} + \frac{1+2+3+4}{4^3} + \cdots.$$

Proof. Let $\sum_{n=1}^{\infty} a_n$ be the given series. Then

$$a_n = \frac{1+2+\cdots+(n+1)}{(n+1)^3} = \frac{(n+1)(n+2)}{2(n+1)^3} = \frac{n+2}{2(n+1)^2}.$$

Choose $b_n = \frac{1}{n}$. Then

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{n(n+2)}{2(n+1)^2} = \frac{1}{2}.$$

Since $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, the given series also diverges by the Limit Comparison Test. □

Example 3.21. Test the convergence of the series

$$\frac{1}{1 \cdot 2^2} + \frac{1}{2 \cdot 3^2} + \frac{1}{3 \cdot 4^2} + \cdots.$$

Proof. Let $\sum_{n=1}^{\infty} a_n$ be the given series. Then $a_n = \frac{1}{n(n+1)^2}$. Choose $b_n = \frac{1}{n^3}$. Then

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{n^2}{(n+1)^2} = 1.$$

Since $\sum_{n=1}^{\infty} \frac{1}{n^3}$ converges, the given series also converges by the Limit Comparison Test. □

Remark 3.22. 1. If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$, then $\sum_{n=1}^{\infty} a_n$ converges whenever $\sum_{n=1}^{\infty} b_n$ converges.

2. If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$, then $\sum_{n=1}^{\infty} a_n$ diverges whenever $\sum_{n=1}^{\infty} b_n$ diverges.

Theorem 3.23 (p -Series Test). The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if $p > 1$ and diverges if $p \leq 1$.

Theorem 3.24 (d'Alembert's Ratio Test). Let $\sum_{n=1}^{\infty} a_n$ be a series of positive real numbers. Suppose that

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = r.$$

Then:

1. If $0 \leq r < 1$, then the series $\sum_{n=1}^{\infty} a_n$ converges.
2. If $r > 1$, then the series $\sum_{n=1}^{\infty} a_n$ diverges.
3. If $r = 1$, then the test is inconclusive.

Example 3.25. 1. Consider the series $\sum_{n=1}^{\infty} \frac{1}{n!}$. Since

$$\frac{a_{n+1}}{a_n} = \frac{1}{n+1} \rightarrow 0,$$

the Ratio Test implies that the series converges.

2. Consider the series $\sum_{n=1}^{\infty} \frac{2^n}{n}$. Since

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1}}{n+1} \cdot \frac{n}{2^n} = 2 \cdot \frac{n}{n+1} \rightarrow 2 > 1,$$

the Ratio Test implies that the series diverges.

3. Consider the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$. Since

$$\frac{a_{n+1}}{a_n} = \frac{n}{n+1} \rightarrow 1,$$

the Ratio Test is inconclusive. Indeed, the harmonic series is divergent. On the other hand, for the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$,

$$\frac{a_{n+1}}{a_n} = \left(\frac{n}{n+1} \right)^2 \rightarrow 1,$$

yet this series is convergent. Thus, no conclusion can be drawn when the limit equals 1.

Theorem 3.26 (Cauchy's Root Test). Let $\sum_{n=1}^{\infty} a_n$ be a series of positive real numbers. Suppose that

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = r.$$

Then:

1. If $0 \leq r < 1$, then the series $\sum_{n=1}^{\infty} a_n$ converges.
2. If $r > 1$, then the series $\sum_{n=1}^{\infty} a_n$ diverges.
3. If $r = 1$, then the test is inconclusive.

Example 3.27. 1. Consider the series $\sum_{n=1}^{\infty} \frac{1}{2^n}$. Since

$$\sqrt[n]{a_n} = \sqrt[n]{\frac{1}{2^n}} = \frac{1}{2},$$

the Root Test implies that the series converges.

2. Consider the series $\sum_{n=1}^{\infty} 2^n$. Since

$$\sqrt[n]{a_n} = \sqrt[n]{2^n} = 2,$$

the Root Test implies that the series diverges.

3. Consider the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$. Since

$$\sqrt[n]{a_n} = \sqrt[n]{\frac{1}{n}} = \frac{1}{n^{1/n}} \rightarrow 1,$$

the Root Test is inconclusive. Indeed, the harmonic series is divergent. On the other hand, for the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$,

$$\sqrt[n]{a_n} = \sqrt[n]{\frac{1}{n^2}} = \frac{1}{n^{2/n}} \rightarrow 1,$$

yet this series is convergent. Thus, no conclusion can be drawn when the limit equals 1.

Example 3.28. Show that the series $\sum_{n=1}^{\infty} \frac{2^n n!}{n^n}$ converges.

Proof. Let $a_n = \frac{2^n n!}{n^n}$. Then

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1}(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{2^n n!} = 2 \left(\frac{n}{n+1} \right)^n.$$

Since $(1 + \frac{1}{n})^n \rightarrow e$, we have

$$\left(\frac{n}{n+1} \right)^n = \frac{1}{(1 + \frac{1}{n})^n} \rightarrow \frac{1}{e}.$$

Hence

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \frac{2}{e} < 1.$$

Therefore, by the Ratio Test, the series $\sum_{n=1}^{\infty} \frac{2^n n!}{n^n}$ converges. □

Theorem 3.29 (Gauss's Test). Let $\sum_{n=1}^{\infty} a_n$ be a series of positive real numbers. Suppose $\frac{a_n}{a_{n+1}} = 1 + \frac{\alpha}{n} + \frac{b_n}{n^p}$, where $p > 1$ and $\{b_n\}$ is a bounded sequence. Then $\sum_{n=1}^{\infty} a_n$ converges if $\alpha > 1$ and diverges if $\alpha \leq 1$.

Example 3.30. Test the convergence of the series

$$1 + \frac{2^2}{3^2} + \frac{2^2 \cdot 4^2}{3^2 \cdot 5^2} + \frac{2^2 \cdot 4^2 \cdot 6^2}{3^2 \cdot 5^2 \cdot 7^2} + \cdots$$

Proof. Let $\sum_{n=1}^{\infty} a_n$ be the given series. For $n \geq 2$, $a_n = \frac{2^2 \cdot 4^2 \cdots (2n-2)^2}{3^2 \cdot 5^2 \cdots (2n-1)^2}$. Then

$$\frac{a_n}{a_{n+1}} = \frac{(2n+1)^2}{(2n)^2} = \frac{4n^2 + 4n + 1}{4n^2} = 1 + \frac{1}{n} + \frac{1}{4n^2}.$$

Thus $\frac{a_n}{a_{n+1}} = 1 + \frac{\alpha}{n} + \frac{b_n}{n^2}$, where $\alpha = 1$ and $b_n = \frac{1}{4}$ is bounded. Hence, by Gauss's Test, the series diverges. $\square$

Definition 3.6. Let $f, \phi : \mathbb{N} \rightarrow \mathbb{R}$, where $\phi(n) > 0$ for all sufficiently large n . We write

$$f(n) = O(\phi(n))$$

if there exist constants $k > 0$ and $N \in \mathbb{N}$ such that

$$|f(n)| \leq k\phi(n), \quad \forall n \geq N.$$

In other words, for sufficiently large n , the function $f(n)$ does not grow faster than a constant multiple of $\phi(n)$. In particular, $O(1)$ denotes the class of all bounded functions.

Example 3.31. Show that the polynomial $f(n) = 5n^2 + 3n + 1$ is of order $O(n^2)$.

Proof. For $n \geq 1$, $3n \leq 3n^2$, $1 \leq n^2$. Hence,

$$f(n) = 5n^2 + 3n + 1 \leq 5n^2 + 3n^2 + n^2 = 9n^2.$$

Thus, $|f(n)| \leq 9n^2$, $\forall n \geq 1$. Therefore $f(n) = O(n^2)$. $\square$

Theorem 3.32 (Alternative Form of Gauss's Test). Let $\sum_{n=1}^{\infty} a_n$ be a series of positive real numbers. If

$$\frac{a_n}{a_{n+1}} = 1 + \frac{\alpha}{n} + O\left(\frac{1}{n^p}\right), \quad p > 1,$$

then the series converges if $\alpha > 1$ and diverges if $\alpha \leq 1$.

Example 3.33. Test the convergence of the series

$$\left(\frac{1}{2}\right)^2 + \left(\frac{1 \cdot 3}{2 \cdot 4}\right)^2 + \left(\frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}\right)^2 + \cdots.$$

Proof. Let $\sum_{n=1}^{\infty} a_n$ be the given series. Then $a_n = \left(\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots 2n}\right)^2$. Hence,

$$\frac{a_n}{a_{n+1}} = \left(\frac{2n+2}{2n+1}\right)^2 = \left(1 + \frac{1}{n}\right)^2 \left(1 + \frac{1}{2n}\right)^{-2}.$$

Using the expansion

$$\left(1 + \frac{1}{2n}\right)^{-2} = 1 - \frac{1}{n} + O\left(\frac{1}{n^2}\right),$$

we obtain $\frac{a_n}{a_{n+1}} = 1 + \frac{1}{n} + O\left(\frac{1}{n^2}\right)$. Therefore, $\alpha = 1$, and by the alternative form of Gauss's Test, the series diverges. $\square$

Exercise 3.2. Test the convergence of the following series.

1.

$$1 + \frac{1}{2}x + \frac{2^2}{3!}x^2 + \frac{3^2}{4!}x^3 + \cdots, \quad x > 0.$$

2.

$$\frac{3}{7} + \frac{3 \cdot 6}{7 \cdot 10}x + \frac{3 \cdot 6 \cdot 9}{7 \cdot 10 \cdot 13}x^2 + \cdots, \quad x > 0.$$

3.

$$\frac{1+x}{1!} + \frac{(1+2x)^2}{2!} + \frac{(1+3x)^3}{3!} + \cdots, \quad x > 0.$$

4.

$$1 + \frac{2^2}{3^2}x + \frac{2^2 \cdot 4^2}{3^2 \cdot 5^2}x^2 + \cdots, \quad x > 0.$$

5.

$$1 + \frac{1}{2!}x + \frac{(2!)^2}{4!}x^2 + \frac{(3!)^2}{6!}x^3 + \cdots, \quad x > 0.$$

6.

$$\frac{a}{b} + \frac{a(a+c)}{b(b+c)} + \frac{a(a+c)(a+2c)}{b(b+c)(b+2c)} + \cdots, \quad a, b, c > 0.$$

3.3 Tests of convergence for Alternating Series

Definition 3.7. A series of the form $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ where (a_n) is a sequence of positive terms is called an alternating series.

Definition 3.8 (Absolutely Convergent Series). An infinite series $\sum_{n=1}^{\infty} a_n$ is said to be **absolutely convergent** if the series

$$\sum_{n=1}^{\infty} |a_n|$$

is convergent.

Theorem 3.34. An absolutely convergent series is convergent.

Proof. Let $\sum_{n=1}^{\infty} u_n$ be a series of positive and negative real numbers and suppose that it is absolutely convergent. Then

$$\sum_{n=1}^{\infty} |u_n|$$

is a convergent series of positive terms.

Let $\varepsilon > 0$ be given. Since $\sum |u_n|$ converges, by Cauchy's criterion there exists a natural number m such that

$$|u_{n+1}| + |u_{n+2}| + \cdots + |u_{n+p}| < \varepsilon$$

for all $n \geq m$ and every natural number p . Now,

$$|u_{n+1} + u_{n+2} + \cdots + u_{n+p}| \leq |u_{n+1}| + |u_{n+2}| + \cdots + |u_{n+p}| < \varepsilon,$$

for all $n \geq m$ and every natural number p . Therefore, by Cauchy's criterion, $\sum_{n=1}^{\infty} u_n$ is convergent. □

Remark 3.35. If $a_n \geq 0$ for every n , then $|a_n| = a_n$, and hence

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} a_n.$$

Therefore, a series of non-negative terms is convergent if and only if it is absolutely convergent.

Example 3.36. 1. The series

$$1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \frac{1}{5^2} - \frac{1}{6^2} + \cdots$$

is convergent since it is absolutely convergent.

2. The series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots$$

is convergent since it is absolutely convergent.

3. For a fixed value of x , the series $\sum_{n=1}^{\infty} \frac{\sin(nx)}{n^2}$ is absolutely convergent. Indeed,

$$\left| \frac{\sin(nx)}{n^2} \right| \leq \frac{1}{n^2},$$

and since $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, the Comparison Test implies that $\sum_{n=1}^{\infty} \left| \frac{\sin(nx)}{n^2} \right|$ also converges.

Hence, $\sum_{n=1}^{\infty} \frac{\sin(nx)}{n^2}$ is absolutely convergent.

Theorem 3.37. If the series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent and $\{b_n\}$ is a bounded sequence, then the series

$$\sum_{n=1}^{\infty} a_n b_n$$

is absolutely convergent.

Example 3.38. Test the convergence of the series

$$\frac{2}{1^3} - \frac{3}{2^3} + \frac{4}{3^3} - \frac{5}{4^3} + \cdots.$$

Proof. Let $\sum_{n=1}^{\infty} a_n$ be the given series. Then

$$a_n = (-1)^{n+1} \frac{n+1}{n^3} = \frac{(-1)^{n+1}}{n^2} \left(1 + \frac{1}{n}\right) = \alpha_n b_n,$$

where, $\alpha_n = \frac{(-1)^{n+1}}{n^2}$, $b_n = 1 + \frac{1}{n}$. Since $\sum_{n=1}^{\infty} \alpha_n$ is absolutely convergent and $\{b_n\}$ is a bounded sequence, it follows from the theorem that $\sum_{n=1}^{\infty} \alpha_n b_n$, that is, the given series, is absolutely convergent. $\square$

Theorem 3.39 (Leibniz Test). *Let (a_n) be a decreasing sequence of non-negative real numbers such that $a_n \rightarrow 0$. Then the alternating series*

$$\sum_{n=1}^{\infty} (-1)^{n-1} a_n$$

converges. Moreover, if $S = \sum_{n=1}^{\infty} (-1)^{n-1} a_n$, then $a_1 - a_2 \leq S \leq a_1$.

Proof. Let $s_n = \sum_{k=1}^n (-1)^{k-1} a_k$. Observe that

$$s_{2n} = (a_1 - a_2) + \cdots + (a_{2n-1} - a_{2n}).$$

Since $a_{2n+1} \geq a_{2n+2}$,

$$s_{2n+2} - s_{2n} = a_{2n+1} - a_{2n+2} \geq 0.$$

Hence (s_{2n}) is increasing. Moreover,

$$s_{2n} = a_1 - (a_2 - a_3) - \cdots - (a_{2n-2} - a_{2n-1}) - a_{2n} \leq a_1.$$

Thus (s_{2n}) is bounded above, and therefore converges. Let $s_{2n} \rightarrow S$. Now, $s_{2n+1} = s_{2n} + a_{2n+1}$. Since $a_{2n+1} \rightarrow 0$, we obtain $s_{2n+1} \rightarrow S$. Hence, both the even and odd subsequences of (s_n) converge to the same limit. Therefore, $s_n \rightarrow S$, which proves that the series converges. Finally,

$$a_1 - a_2 = s_2 \leq s_{2n} \leq S \leq s_{2n+1} \leq s_1 = a_1,$$

and letting $n \rightarrow \infty$ gives $a_1 - a_2 \leq S \leq a_1$. $\square$

Example 3.40. *Consider the alternating harmonic series*

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}.$$

Since the sequence $a_n = \frac{1}{n}$ is decreasing and satisfies $a_n \rightarrow 0$, the Leibniz Test implies that the series converges.

However, the series is not absolutely convergent, because

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^{n-1}}{n} \right| = \sum_{n=1}^{\infty} \frac{1}{n},$$

which is the harmonic series and is divergent (see Example 19). Thus, the alternating harmonic series is conditionally convergent.

Example 3.41. 1. The alternating harmonic series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$$

is convergent by the Leibniz Alternating Series Test.

2. For $a > 0$, the series

$$\frac{1}{1+a^2} - \frac{1}{2+a^2} + \frac{1}{3+a^2} - \cdots$$

is convergent by the Leibniz Alternating Series Test.

3. The series

$$1 - \frac{1}{2} + \frac{1 \cdot 3}{2 \cdot 4} - \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} + \cdots$$

is convergent by the Leibniz Alternating Series Test, since

$$\lim_{n \rightarrow \infty} \frac{1 \cdot 3 \cdots (2n-1)}{2 \cdot 4 \cdots (2n)} = 0,$$

and

$$\frac{1 \cdot 3 \cdots (2n-1)}{2 \cdot 4 \cdots (2n)} > \frac{1 \cdot 3 \cdots (2n+1)}{2 \cdot 4 \cdots (2n+2)},$$

so the terms decrease monotonically to zero.

Theorem 3.42 (Abel's Test). Let $\sum a_n$ be a convergent series, and let $\{b_n\}$ be a monotone bounded sequence. Then the series

$$\sum_{n=1}^{\infty} a_n b_n$$

is convergent.

Example 3.43. 1. The series

$$\sum_{n=2}^{\infty} \frac{(-1)^{n+1}}{n \log n}$$

is convergent by Abel's Test, since $\sum_{n=2}^{\infty} \frac{(-1)^{n+1}}{n}$ is convergent and the sequence $\left\{ \frac{1}{\log n} \right\}_{n=2}^{\infty}$ is monotone decreasing and bounded below.

2. The series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n+1} \left(1 + \frac{1}{n} \right)^{-n}$$

is convergent by Abel's Test, since $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n+1}$ is convergent and the sequence $\left\{ \left(1 + \frac{1}{n} \right)^{-n} \right\}_{n=1}^{\infty}$ is monotone decreasing and bounded below.

Definition 3.9 (Conditionally Convergent Series). A series is said to be **conditionally convergent** if it is convergent but not absolutely convergent.

Example 3.44. Show that the series

$$\frac{1}{(1+a)^p} - \frac{1}{(2+a)^p} + \frac{1}{(3+a)^p} - \cdots, \quad a > 0$$

is

1. absolutely convergent if $p > 1$,
2. conditionally convergent if $0 < p \leq 1$.

Proof. Let $\sum_{n=1}^{\infty} a_n$ be the given series and let $b_n = |a_n|$. Then $\sum_{n=1}^{\infty} b_n$ is a series of positive real numbers and $b_n = \frac{1}{(n+a)^p}$. Let

$$c_n = \frac{1}{n^p}.$$

Then

$$\lim_{n \rightarrow \infty} \frac{b_n}{c_n} = \lim_{n \rightarrow \infty} \frac{n^p}{(n+a)^p} = 1.$$

By the Limit Comparison Test, $\sum_{n=1}^{\infty} b_n$ converges if $p > 1$ and diverges if $0 < p \leq 1$.

Case 1: $p > 1$.

In this case, $\sum_{n=1}^{\infty} b_n$ converges. Hence $\sum_{n=1}^{\infty} a_n$ is absolutely convergent.

Case 2: $0 < p \leq 1$.

In this case, $b_n = \frac{1}{(n+a)^p}$ is a monotone decreasing sequence of positive real numbers and $\lim_{n \rightarrow \infty} b_n = 0$. Since the given series is alternating, by Leibniz's Test, $\sum_{n=1}^{\infty} a_n$ converges. But

$\sum_{n=1}^{\infty} b_n$ diverges. Therefore, $\sum_{n=1}^{\infty} a_n$ is conditionally convergent. □

Exercise 3.3. Test the convergence of each of the following series.

(i)

$$1 - \frac{1}{2!} + \frac{1}{4!} - \frac{1}{6!} + \cdots.$$

(ii)

$$\frac{1}{2^2 \log 2} - \frac{1}{3^2 \log 3} + \frac{1}{4^2 \log 4} - \cdots.$$

(iii)

$$\frac{x}{1+x} - \frac{x^2}{1+x^2} + \frac{x^3}{1+x^3} - \cdots, \quad 0 < x < 1.$$

(iv)

$$1 - \frac{1}{2} + \frac{1 \cdot 3}{2 \cdot 4} - \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} + \cdots.$$

(v)

$$\frac{1}{(\log 2)^2} - \frac{1}{(\log 3)^2} + \frac{1}{(\log 4)^2} - \cdots.$$